

# DIT Global Public Health — Assessment Test Question Bank

165 questions across 4 categories · Short answer + full explanation

| Category 1 – General Knowledge |                                                                             |                          |                                                                                                                                                                                                                                                                                                                             |
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| #                              | Question                                                                    | Answer                   | Explanation                                                                                                                                                                                                                                                                                                                 |
| 1                              | What is the largest ocean in the world?                                     | <b>Pacific Ocean</b>     | The Pacific Ocean is the largest ocean on Earth, covering about 165 million km <sup>2</sup> — more than all the world's landmasses combined. It spans from the Arctic in the north to the Antarctic in the south. The Southern Ocean (around Antarctica) is sometimes listed separately, making the Atlantic third-largest. |
| 2                              | What is the longest river in the world?                                     | <b>Nile</b>              | The Nile River (~6,650 km) is traditionally regarded as the world's longest river, flowing northward through northeastern Africa into the Mediterranean Sea. The Amazon is sometimes cited as longer depending on the measurement method used — making this a genuinely contested fact.                                     |
| 3                              | Which is the largest continent by area?                                     | <b>Asia</b>              | Asia is the largest continent by area (~44.6 million km <sup>2</sup> ) and also the most populous, home to over 60% of the world's population. Africa is the second-largest continent (~30.3 million km <sup>2</sup> ) — a common wrong answer.                                                                             |
| 4                              | Which is the largest desert in the world (overall, including cold deserts)? | <b>Antarctica</b>        | Antarctica is the largest desert overall (~14 million km <sup>2</sup> ) because a desert is defined by low precipitation, not heat. The Sahara is the largest hot desert (~9 million km <sup>2</sup> ).                                                                                                                     |
| 5                              | What is the tallest mountain in the world?                                  | <b>Mount Everest</b>     | Mount Everest (8,849 m above sea level) is the world's tallest mountain, located in the Himalayas on the Nepal–China border. K2 (8,611 m) is the second-highest and often considered the most technically difficult to climb.                                                                                               |
| 6                              | What is the capital of Australia?                                           | <b>Canberra</b>          | Canberra is the capital of Australia — a common trick question since Sydney and Melbourne are larger cities but were bypassed when the capital was established in 1913.                                                                                                                                                     |
| 7                              | Which city serves as the administrative/executive capital of South Africa?  | <b>Pretoria</b>          | South Africa has three capitals: Pretoria (administrative/executive), Cape Town (legislative/parliamentary), and Bloemfontein (judicial). The question specifically asks about the administrative capital, which is Pretoria.                                                                                               |
| 8                              | Who discovered penicillin?                                                  | <b>Alexander Fleming</b> | Alexander Fleming discovered penicillin in 1928 when he noticed that the mould <i>Penicillium notatum</i> was killing bacteria on a culture plate. This discovery revolutionized medicine and led to the antibiotic era.                                                                                                    |
| 9                              | What is the boiling point of water at sea level?                            | <b>100°C</b>             | Water boils at 100°C (212°F) at standard atmospheric pressure (1 atm / sea level). At higher altitudes where atmospheric pressure is lower, water boils at a lower temperature — relevant to food safety and sterilisation in high-altitude settings.                                                                       |

| #  | Question                                                            | Answer           | Explanation                                                                                                                                                                                                                                                                                                      |
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| 10 | What is the freezing point of water at standard pressure?           | 0°C              | Water freezes at 0°C (32°F) at standard atmospheric pressure. Note: 4°C is the temperature at which water reaches maximum density — a common confusion. Below 0°C, water transitions from liquid to solid (ice).                                                                                                 |
| 11 | What is the approximate speed of light in a vacuum?                 | 300,000 km/s     | The speed of light in a vacuum is approximately 299,792 km/s, commonly rounded to 300,000 km/s. According to Einstein's theory of relativity, nothing in the universe can travel faster than light.                                                                                                              |
| 12 | Which planet has the highest number of confirmed moons?             | Saturn           | As of 2024, Saturn has the most confirmed moons of any planet — 146 — surpassing Jupiter. Saturn's moons vary greatly in size, from tiny moonlets to the large Titan.                                                                                                                                            |
| 13 | Which planet is closest to the Sun?                                 | Mercury          | Mercury is the closest planet to the Sun, orbiting at an average distance of ~58 million km. Despite being closest to the Sun, Mercury is not the hottest planet — Venus is, due to its thick CO <sub>2</sub> atmosphere trapping heat (greenhouse effect).                                                      |
| 14 | Which element has the chemical symbol 'O'?                          | Oxygen           | Oxygen has the chemical symbol 'O' and atomic number 8. It makes up approximately 21% of Earth's atmosphere and is essential for cellular respiration. Note: Ozone (O <sub>3</sub> ) is a molecule made of three oxygen atoms, not an element with its own symbol.                                               |
| 15 | What is the hardest natural substance on Earth?                     | Diamond          | Diamond is the hardest natural substance on Earth, rating 10 (maximum) on the Mohs hardness scale. It is made of pure carbon in a cubic crystal structure. Corundum (9) and topaz (8) are the next hardest — all are plausible options.                                                                          |
| 16 | Which gas do plants absorb during photosynthesis?                   | Carbon dioxide   | Plants absorb carbon dioxide (CO <sub>2</sub> ) during photosynthesis and combine it with water and sunlight to produce glucose (energy) and oxygen as a by-product. This process is why plants are called carbon sinks — relevant to climate change and public health discussions.                              |
| 17 | What is the chemical formula for water?                             | H <sub>2</sub> O | The chemical formula for water is H <sub>2</sub> O — two hydrogen atoms covalently bonded to one oxygen atom. H <sub>2</sub> O <sub>2</sub> is hydrogen peroxide (a disinfectant), which is a common confusion. Water's unique polarity makes it the 'universal solvent' essential for all biological processes. |
| 18 | Who developed the theory of relativity?                             | Albert Einstein  | Albert Einstein developed Special Relativity (1905), which introduced E=mc <sup>2</sup> , and General Relativity (1915), which described gravity as the curvature of spacetime. Newton developed classical mechanics; Planck developed quantum theory; Bohr developed the atomic model.                          |
| 19 | Which country has the largest population in the world (as of 2023)? | India            | India surpassed China in 2023 and now holds the top position with over 1.4 billion people, making it the world's most populous country.                                                                                                                                                                          |
| 20 | Which country is the largest by land area?                          | Russia           | Russia is the largest country by land area, covering approximately 17.1 million km <sup>2</sup> across 11 time zones. Canada is the second-largest (~10 million km <sup>2</sup> ), a common wrong answer. Russia spans both Europe and Asia.                                                                     |

| #  | Question                                                                                                   | Answer                                                        | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 21 | What is the most widely spoken native language in the world?                                               | <b>Mandarin Chinese</b>                                       | Mandarin Chinese has the most native speakers (~1 billion), making it the most widely spoken native language. English is the most widely spoken language overall when including second-language speakers (~1.5 billion total). Hindi is the third most spoken native language. This is a commonly confused question — native speakers vs total speakers give different rankings.                                                              |
| 22 | In which year did World War II end?                                                                        | <b>1945</b>                                                   | World War II ended in 1945 — Germany surrendered in May (V-E Day) and Japan surrendered in September (V-J Day) following the atomic bombings of Hiroshima and Nagasaki. The post-war period directly led to the founding of the United Nations (1945) and the WHO (1948).                                                                                                                                                                     |
| 23 | What is the currency of the European Union?                                                                | <b>Euro</b>                                                   | The Euro (€) is the currency of the Eurozone, currently used by 20 of the 27 EU member states. Some EU members (e.g. Sweden, Poland) retain their own national currencies.                                                                                                                                                                                                                                                                    |
| 24 | Which blood type is the universal donor for red blood cell transfusions?                                   | <b>O negative</b>                                             | O negative (O-) is the universal donor for red blood cell transfusions because it lacks A antigens, B antigens, and the Rh factor — making it compatible with any blood type in emergency situations.                                                                                                                                                                                                                                         |
| 25 | What is the approximate world population as of 2024?                                                       | <b>8.1 billion</b>                                            | The world population reached 8 billion in November 2022 and stands at approximately 8.1 billion as of 2024, with growth concentrated in sub-Saharan Africa and South Asia.                                                                                                                                                                                                                                                                    |
| 26 | In which country is the WHO headquarters located?                                                          | <b>Switzerland (Geneva)</b>                                   | The WHO is headquartered in Geneva, Switzerland. It was established in 1948 as a specialised agency of the United Nations responsible for international public health.                                                                                                                                                                                                                                                                        |
| 27 | Which UN body is responsible for setting global health policy?                                             | <b>World Health Organization (WHO)</b>                        | The World Health Organization (WHO) is the UN body responsible for directing and coordinating international health work. It has 194 member states and is headquartered in Geneva.                                                                                                                                                                                                                                                             |
| 28 | What does UNICEF stand for?                                                                                | <b>United Nations International Children's Emergency Fund</b> | UNICEF stands for United Nations International Children's Emergency Fund (now officially just 'UNICEF'). It operates in 190+ countries to protect children's rights, health, nutrition, and education.                                                                                                                                                                                                                                        |
| 29 | Which of the following is a permanent member of the UN Security Council?                                   | <b>France</b>                                                 | The five permanent members (P5) of the UN Security Council — USA, UK, France, Russia, and China — each hold veto power over Security Council resolutions.                                                                                                                                                                                                                                                                                     |
| 30 | Which international agreement, adopted in 2015, set binding targets for reducing greenhouse gas emissions? | <b>Paris Agreement</b>                                        | The Paris Agreement (2015) is the landmark international climate treaty adopted under the UNFCCC at COP21. It sets the goal of limiting global warming to well below 2°C above pre-industrial levels, with efforts to limit it to 1.5°C. Unlike the Kyoto Protocol (1997), it is universal — applying to all countries, not just developed ones. The Montreal Protocol (1987) addressed ozone depletion, not greenhouse gases.                |
| 31 | The concept of 'One Health' recognizes the interconnection between which three domains?                    | <b>Human health, animal health, and environmental health</b>  | One Health is an integrated approach recognising that human health, animal health, and environmental health are deeply interconnected and interdependent. Most emerging infectious diseases (e.g. Ebola, COVID-19, avian influenza) originate at the human-animal-environment interface (zoonotic spillover). The WHO, FAO, and UNEP jointly promote the One Health approach as essential for pandemic prevention and global health security. |

| #  | Question                                                                              | Answer                                                                                              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 32 | Which of the following best describes the 'demographic transition'?                   | <b>The shift from high birth and death rates to low birth and death rates as a country develops</b> | The demographic transition model describes how countries move through four stages: Stage 1 — high birth rate, high death rate (pre-industrial); Stage 2 — death rate falls (improved sanitation/food), birth rate stays high → population boom; Stage 3 — birth rate begins to fall; Stage 4 — low birth rate, low death rate (post-industrial). This transition has major implications for healthcare burden — ageing populations in Stage 4 countries shift disease burden toward non-communicable diseases (NCDs).                                                      |
| 33 | What does 'TRIPS' stand for, and why is it relevant to global public health?          | <b>Trade-Related Intellectual Property Rights; it affects access to affordable medicines</b>        | TRIPS (Trade-Related Aspects of Intellectual Property Rights) is a WTO agreement that sets minimum standards for intellectual property protection, including pharmaceutical patents. It is a major public health issue because it can restrict access to affordable generic medicines in low- and middle-income countries. The Doha Declaration (2001) clarified that TRIPS should not prevent countries from taking measures to protect public health, including issuing compulsory licences for generic production of patented medicines (e.g. antiretrovirals for HIV). |
| 34 | Which country has the highest under-5 mortality rate globally, as of recent WHO data? | <b>Sierra Leone</b>                                                                                 | Sierra Leone consistently ranks among the highest under-5 mortality rates globally (approximately 100-105 per 1,000 live births), driven by high rates of malaria, pneumonia, diarrhoeal disease, neonatal complications, and severely limited healthcare infrastructure. Chad and Nigeria also rank very high. This contrasts with countries like Finland and Japan where under-5 mortality is below 3 per 1,000. Under-5 mortality is a key indicator of a country's overall health system performance and child health status.                                          |

## Category 2 – Body Systems, Physiology & Public Health Biology

| #  | Question                                                                                     | Answer                                                                                | Explanation                                                                                                                                                                                                                                                                                                           |
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| 1  | What is the primary function of the skin?                                                    | <b>Protection, temperature regulation, sensation, and vitamin D synthesis</b>         | The skin has four primary functions: (1) protection — barrier against pathogens, UV radiation, and injury; (2) temperature regulation — via sweating and vasodilation/vasoconstriction; (3) sensation — touch, pain, temperature receptors; (4) Vitamin D synthesis — UV-B triggers production of Vitamin D■.         |
| 2  | True or False: The nervous system is the ONLY system responsible for temperature regulation. | <b>False – it also involves the integumentary, endocrine, and circulatory systems</b> | FALSE. While the nervous system plays a key role (hypothalamus as thermostat), temperature regulation also involves: the integumentary system (sweating), the circulatory system (vasodilation/vasoconstriction), and the endocrine system (thyroid hormones affecting metabolic rate). It is a multi-system process. |
| 3  | Which organ is responsible for filtering blood and producing urine?                          | <b>Kidney</b>                                                                         | The kidneys are the primary organs for blood filtration and urine production. They filter approximately 180 litres of blood per day, reabsorbing useful substances and excreting waste products (urea, creatinine) as urine via the ureters.                                                                          |
| 4  | Which part of the brain controls balance and coordination?                                   | <b>Cerebellum</b>                                                                     | The cerebellum, located at the posterior base of the brain, is responsible for coordinating voluntary movements, maintaining balance and posture, and regulating fine motor control. Damage causes ataxia (uncoordinated movement).                                                                                   |
| 5  | What is the normal resting heart rate in adults?                                             | <b>60–100 bpm</b>                                                                     | The normal resting heart rate in adults is 60–100 beats per minute (bpm). Below 60 bpm is bradycardia (normal in athletes); above 100 bpm is tachycardia.                                                                                                                                                             |
| 6  | What is the primary function of the mitochondria?                                            | <b>Energy (ATP) production via oxidative phosphorylation</b>                          | Mitochondria are the 'powerhouses of the cell' — they generate ATP (adenosine triphosphate) through aerobic respiration (oxidative phosphorylation). They also regulate apoptosis (programmed cell death) and contain their own DNA.                                                                                  |
| 7  | Which chamber of the heart pumps oxygenated blood to the body?                               | <b>Left ventricle</b>                                                                 | The left ventricle is the most muscular chamber of the heart. It pumps oxygenated blood into the aorta and out to the entire body (systemic circulation). The right ventricle pumps deoxygenated blood to the lungs.                                                                                                  |
| 8  | What is the normal body temperature in humans?                                               | <b>37.0°C</b>                                                                         | Normal human body temperature is 37°C (98.6°F), with a physiological range of 36.1–37.2°C. A temperature above 38°C is generally defined as fever (pyrexia).                                                                                                                                                          |
| 9  | Which hormone lowers blood sugar levels?                                                     | <b>Insulin</b>                                                                        | Insulin (produced by beta cells of the pancreatic islets of Langerhans) lowers blood glucose by facilitating cellular uptake. Glucagon (from alpha cells) raises blood glucose by stimulating glycogen breakdown. Together they maintain glucose homeostasis.                                                         |
| 10 | What is the primary role of white blood cells?                                               | <b>Immune defense against pathogens</b>                                               | White blood cells (leukocytes) form the body's immune defence. Key types: neutrophils (kill bacteria), B-cells (produce antibodies), T-cells (viral immunity and coordination), macrophages (engulf pathogens). They circulate in blood and lymph.                                                                    |

| #  | Question                                                                                | Answer                                                                                              | Explanation                                                                                                                                                                                                                                                                                                                                |
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| 11 | Which part of the cell contains genetic material (DNA)?                                 | <b>Nucleus</b>                                                                                      | The nucleus contains most of the cell's genetic material — DNA organised into 46 chromosomes (23 pairs) in humans. It controls gene expression and directs cell division (mitosis/meiosis).                                                                                                                                                |
| 12 | What is the function of the hypothalamus?                                               | <b>Regulates temperature, hunger, thirst, and pituitary hormone release</b>                         | The hypothalamus is the brain's homeostatic control centre. It regulates body temperature, hunger, thirst, sleep-wake cycles, and emotional responses, and controls the pituitary gland (via releasing/inhibiting hormones), making it central to the endocrine system.                                                                    |
| 13 | True or False: Neurons can fully regenerate after injury in the central nervous system. | <b>False – CNS neurons have very limited regenerative capacity</b>                                  | FALSE. Central nervous system (CNS) neurons have very limited capacity to regenerate after injury, due to inhibitory factors in the CNS environment (e.g. myelin-associated inhibitors like Nogo). Peripheral nervous system (PNS) neurons can regenerate partially if the cell body is intact.                                            |
| 14 | Which blood cells are primarily involved in clotting?                                   | <b>Platelets (thrombocytes)</b>                                                                     | Platelets (thrombocytes) are the blood cells involved in clotting (haemostasis). When a vessel is injured, platelets aggregate at the site and release clotting factors that trigger a cascade forming a fibrin clot to stop bleeding.                                                                                                     |
| 15 | What is the primary function of the lymphatic system?                                   | <b>Immune defense and fluid balance</b>                                                             | The lymphatic system has three main functions: (1) drains excess tissue fluid (lymph) back into the bloodstream, preventing oedema; (2) absorbs dietary fats from the gut via lacteals; (3) houses lymphocytes (T and B cells) for immune surveillance and response.                                                                       |
| 16 | Which gland is known as the 'master gland'?                                             | <b>Pituitary gland</b>                                                                              | The pituitary gland (hypophysis) is called the 'master gland' because it secretes hormones that regulate virtually all other endocrine glands — including the thyroid (TSH), adrenal glands (ACTH), and gonads (LH, FSH). It is itself controlled by the hypothalamus.                                                                     |
| 17 | True or False: The pancreas has both endocrine and exocrine functions.                  | <b>True – it secretes digestive enzymes (exocrine) and hormones like insulin (endocrine)</b>        | TRUE. The pancreas has dual functions: Exocrine — acinar cells secrete digestive enzymes (lipase, amylase, proteases) into the duodenum via the pancreatic duct. Endocrine — islets of Langerhans secrete insulin and glucagon directly into the bloodstream.                                                                              |
| 18 | What is the normal pH of human blood?                                                   | <b>7.35–7.45</b>                                                                                    | Normal human blood pH is 7.35–7.45 (slightly alkaline). Values below 7.35 indicate acidosis; above 7.45 indicate alkalosis. Both are life-threatening if severe and are tightly regulated by the lungs (CO <sub>2</sub> ) and kidneys (bicarbonate).                                                                                       |
| 19 | Which vitamin deficiency causes scurvy?                                                 | <b>Vitamin C (scurvy)</b>                                                                           | Vitamin C (ascorbic acid) deficiency causes scurvy. Vitamin C is essential for collagen synthesis (hydroxylation of proline and lysine). Deficiency leads to bleeding gums, poor wound healing, perifollicular haemorrhages, and eventually death if untreated.                                                                            |
| 20 | What is the key difference between innate and adaptive immunity?                        | <b>Innate is non-specific and immediate; adaptive is specific and develops immunological memory</b> | Innate immunity is the body's immediate, non-specific first line of defence (skin, macrophages, inflammation, complement system) — present from birth. Adaptive immunity is slower (days to weeks), antigen-specific, and develops immunological memory through T and B lymphocytes — allowing a faster, stronger response on re-exposure. |

| #  | Question                                                                         | Answer                                                                                                                   | Explanation                                                                                                                                                                                                                                                                                                                                                                                               |
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| 21 | What is herd immunity?                                                           | <b>Indirect protection when sufficient population immunity blocks disease spread</b>                                     | Herd immunity occurs when a sufficient proportion of a population is immune (through vaccination or prior infection) so that an infectious disease cannot spread easily, indirectly protecting non-immune individuals. The required threshold varies by disease (e.g. ~95% for measles, ~80–85% for polio).                                                                                               |
| 22 | True or False: ALL contraceptive methods act on the endocrine system.            | <b>False – barrier methods (e.g., condoms) work mechanically, not hormonally</b>                                         | FALSE. Hormonal contraceptives (combined pill, progestogen-only pill, implant, injection, hormonal IUD) act on the endocrine system. However, barrier methods (male/female condoms, diaphragm) work mechanically to block sperm — they do not involve the endocrine system at all. Therefore NOT all contraceptive methods act on the endocrine system.                                                   |
| 23 | What is the key difference between active and passive immunity?                  | <b>Active immunity involves the host generating its own antibodies; passive involves receiving ready-made antibodies</b> | Active immunity: the body produces its own antibodies in response to an antigen — either from natural infection or vaccination. It is long-lasting and creates immunological memory. Passive immunity: ready-made antibodies are transferred from another source (e.g. maternal antibodies via placenta or breast milk, or immunoglobulin therapy). It is immediate but temporary as no memory is formed. |
| 24 | Which type of immunity is acquired through vaccination?                          | <b>Active artificial immunity</b>                                                                                        | Vaccination produces active artificial immunity — 'active' because the body generates its own antibodies and memory cells, 'artificial' because it is achieved via a vaccine rather than natural infection. This is distinct from passive immunity (no antibody production by the host).                                                                                                                  |
| 25 | What does a vaccine contain that triggers an immune response?                    | <b>Antigens (killed/weakened pathogens or proteins) that stimulate antibody production</b>                               | Vaccines contain antigens — substances that trigger an immune response. These may be killed or live-attenuated (weakened) pathogens, subunit proteins, toxoids (inactivated toxins), or mRNA encoding pathogen proteins (e.g. COVID-19 mRNA vaccines). The immune system responds by producing antibodies and memory cells.                                                                               |
| 26 | Which of the following are the two main types of malnutrition?                   | <b>Undernutrition and overnutrition</b>                                                                                  | The two main types of malnutrition are: (1) Undernutrition — includes protein-energy malnutrition (kwashiorkor, marasmus) and micronutrient deficiencies (iron, vitamin A, iodine). (2) Overnutrition — excess caloric intake leading to overweight and obesity. Both coexist in many LMICs ('double burden of malnutrition').                                                                            |
| 27 | What is kwashiorkor?                                                             | <b>Severe protein deficiency presenting with oedema, distended abdomen, and skin/hair changes</b>                        | Kwashiorkor is severe protein deficiency, typically seen in children after weaning onto a low-protein diet while caloric intake may be adequate. Classic features: pitting oedema (due to hypoalbuminaemia), distended abdomen, skin lesions, and hair depigmentation/loss. Distinguished from marasmus (overall caloric deficiency).                                                                     |
| 28 | What is the leading preventable cause of maternal death in low-income countries? | <b>Postpartum haemorrhage (PPH)</b>                                                                                      | Postpartum haemorrhage (PPH) — defined as blood loss >500 mL after vaginal delivery — is the leading preventable cause of maternal mortality globally, accounting for ~27% of maternal deaths. Most deaths are preventable with skilled birth attendance, active management of the third stage of labour, and oxytocin.                                                                                   |
| 29 | Which intervention has the greatest impact on reducing maternal mortality?       | <b>Skilled birth attendance and access to emergency obstetric care</b>                                                   | The most impactful interventions for reducing maternal mortality are skilled birth attendance and access to emergency obstetric care (EmOC). Additional key interventions include: antenatal care (ANC), iron and folate supplementation to prevent anaemia, family planning, and clean delivery practices.                                                                                               |



| #  | Question                                                                                                                                                    | Answer                                                                                                     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| 30 | What does 'vector-borne' disease transmission mean?                                                                                                         | <b>Disease transmitted via a living organism such as a mosquito or tick</b>                                | Vector-borne transmission means disease is transmitted to humans through a living organism (vector) — typically an arthropod. Examples: malaria and dengue via Anopheles/Aedes mosquitoes; Lyme disease via ticks; sleeping sickness (trypanosomiasis) via tsetse flies. The vector may be a biological host or merely a mechanical carrier.                                                                                                                                                                                                                                                                                                                                     |
| 31 | A patient presents with confusion, rapid breathing, and blood pH of 7.28. What is the most likely diagnosis?                                                | <b>Metabolic acidosis</b>                                                                                  | Normal blood pH is 7.35–7.45. A pH of 7.28 indicates acidosis (below 7.35). Rapid breathing (Kussmaul breathing) is a compensatory response where the respiratory system tries to blow off CO <sub>2</sub> to raise pH — this is the hallmark of METABOLIC acidosis (e.g. diabetic ketoacidosis, lactic acidosis, renal failure). Respiratory acidosis would show slow/shallow breathing with CO <sub>2</sub> retention. Alkalosis (metabolic or respiratory) would show pH above 7.45. This clinical reasoning is essential for public health emergencies and critical care contexts.                                                                                           |
| 32 | In which part of the nephron does the majority of sodium and water reabsorption occur?                                                                      | <b>Proximal convoluted tubule</b>                                                                          | Approximately 65-70% of filtered sodium and water is reabsorbed in the proximal convoluted tubule (PCT) — the largest proportion of any nephron segment. The loop of Henle creates an osmotic gradient important for concentrating urine. The distal convoluted tubule and collecting duct are responsible for fine-tuning under hormonal control (aldosterone controls Na <sup>+</sup> reabsorption; ADH controls water reabsorption). Understanding renal physiology is important for understanding conditions like hypertension, oedema, and diuretic drug mechanisms.                                                                                                        |
| 33 | A child in a refugee camp presents with oedema, a distended abdomen, and normal weight-for-height. Which nutritional deficiency best explains this picture? | <b>Protein deficiency (kwashiorkor) — low albumin causing oedema and distended abdomen</b>                 | The combination of oedema + distended abdomen + relatively preserved weight-for-height is classic kwashiorkor — severe PROTEIN deficiency. Oedema results from hypoalbuminaemia (low albumin reduces oncotic pressure, causing fluid to leak into tissues). Marasmus (total caloric deficiency) presents with severe wasting, no oedema, and very low weight-for-height — the child looks 'skin and bones'. This distinction is critical in humanitarian settings: kwashiorkor requires protein supplementation, while marasmus requires overall caloric rehabilitation. Both can coexist (marasmic-kwashiorkor).                                                                |
| 34 | Which mechanism explains why people with sickle cell trait (HbAS) have relative protection against severe malaria?                                          | <b>Infected red cells sickle and are cleared more rapidly by the spleen, limiting parasite replication</b> | This is a classic example of natural selection and gene-disease interaction. In sickle cell TRAIT (HbAS — one normal, one sickle gene), red blood cells infected by Plasmodium falciparum sickle more readily under the low-oxygen conditions inside infected cells. These sickled infected cells are recognised and cleared by the spleen before the parasite can complete its life cycle and replicate widely. This confers significant protection against severe (cerebral) malaria. This is why HbS allele frequency is high in malaria-endemic regions of sub-Saharan Africa — a textbook example of balanced polymorphism and evolutionary pressure on disease resistance. |
| 35 | True or False: Herd immunity through vaccination can protect individuals who are immunocompromised and cannot receive live vaccines.                        | <b>True</b>                                                                                                | TRUE. This is precisely one of the most important reasons for achieving high vaccination coverage. Immunocompromised individuals (e.g. people on chemotherapy, those with HIV/AIDS, neonates, transplant recipients) cannot receive live attenuated vaccines (e.g. MMR, BCG, yellow fever) due to risk of vaccine-strain infection. When enough of the surrounding population is vaccinated (herd immunity threshold), the pathogen cannot spread efficiently, indirectly protecting those who cannot be vaccinated. This is why measles outbreaks occur in communities with even modest drops in coverage below the 95% threshold.                                              |





### Category 3 – Public Health: Epidemiology & Study Designs

| # | Question                                           | Answer                                                                                                     | Explanation                                                                                                                                                                                                                                                                                                                                                |
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| 1 | What does 'incidence' measure in epidemiology?     | <b>Number of new cases arising in a population over a specific time period</b>                             | Incidence measures the number of NEW cases of a disease arising in a defined population over a specified time period. It reflects the RISK of developing the disease. Formula: Incidence rate = (New cases ÷ Population at risk) × 1,000 (or 100,000). Contrast with prevalence (all existing cases).                                                      |
| 2 | What does 'prevalence' measure in epidemiology?    | <b>Total number of existing cases (new and pre-existing) in a population at a given time</b>               | Prevalence is the total number of EXISTING cases (both new and pre-existing) in a population at a given time. It reflects the overall burden of disease. Formula: Prevalence = (All existing cases ÷ Total population) × 1,000. High prevalence can reflect either high incidence or long disease duration (or both).                                      |
| 3 | What does 'mortality rate' measure?                | <b>The number of deaths in a population over a specific time period</b>                                    | Mortality rate measures the number of deaths in a population over a specific time period, usually expressed per 1,000 or 100,000 people per year. It can refer to all-cause mortality or cause-specific mortality (e.g. maternal mortality rate, infant mortality rate).                                                                                   |
| 4 | What is epidemiology?                              | <b>The study of distribution and determinants of health-related states in populations</b>                  | Epidemiology is the study of the distribution (who, where, when) and determinants (why, how) of health-related states and events in populations, and the application of this knowledge to prevent and control health problems. It is the core science of public health.                                                                                    |
| 5 | What does 'sensitivity' of a diagnostic test mean? | <b>The proportion of people with the disease who test positive (true positive rate)</b>                    | Sensitivity = True Positive Rate — the proportion of people WITH the disease who test positive. A highly sensitive test has few false negatives. Mnemonic: SnNout — a highly Sensitive test, when Negative, rules OUT the disease. Best used for screening.                                                                                                |
| 6 | What does 'specificity' of a diagnostic test mean? | <b>The proportion of people without the disease who test negative (true negative rate)</b>                 | Specificity = True Negative Rate — the proportion of people WITHOUT the disease who test negative. A highly specific test has few false positives. Mnemonic: SpPin — a highly Specific test, when Positive, rules IN the disease. Best used for confirmation/diagnosis.                                                                                    |
| 7 | What is a DALY (Disability-Adjusted Life Year)?    | <b>A measure of disease burden combining years lost to premature death and years lived with disability</b> | A DALY (Disability-Adjusted Life Year) represents one year of healthy life lost. It combines: (1) Years of Life Lost (YLL) — due to premature death; and (2) Years Lived with Disability (YLD) — due to illness/disability. DALY = YLL + YLD. Used by WHO to measure the global burden of disease and compare health loss across conditions and countries. |
| 8 | What is 'attributable risk'?                       | <b>The proportion of disease in the exposed group attributable to the exposure</b>                         | Attributable risk (risk difference) = Incidence in exposed group – Incidence in unexposed group. It quantifies how much of the disease risk in the exposed group is due to the exposure itself. Useful for public health interventions — tells you how many cases could be prevented by removing the exposure.                                             |
| 9 | What is a randomised controlled trial (RCT)?       | <b>An experimental study where participants are randomly assigned to intervention or control</b>           | An RCT is the gold standard for establishing causality. Participants are randomly allocated to an intervention or control group, minimising confounding. It allows determination of whether a treatment actually causes the observed outcome. Limitations: expensive, ethical constraints, not always feasible.                                            |

| #  | Question                                                         | Answer                                                                                                             | Explanation                                                                                                                                                                                                                                                                                                                                                 |
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| 10 | What is a cohort study?                                          | <b>A longitudinal study following an exposed and unexposed group over time to compare outcomes</b>                 | A cohort study follows a defined group (cohort) over time, comparing those exposed to a risk factor versus unexposed, and measuring who develops disease. It directly measures INCIDENCE and is suited to studying multiple outcomes from one exposure. Can be prospective (forward in time) or retrospective (using historical data).                      |
| 11 | What is a case-control study?                                    | <b>A study selecting cases with a disease and controls without, to compare prior exposures</b>                     | A case-control study identifies people who already have the disease (cases) and compares them to people without the disease (controls), then looks backward to compare exposure histories. Efficient for rare diseases and quick to conduct. It estimates the odds ratio (OR) but cannot directly calculate incidence.                                      |
| 12 | Which study design provides the highest level of evidence?       | <b>Systematic review / meta-analysis</b>                                                                           | Systematic reviews and meta-analyses sit at the top of the evidence hierarchy because they pool results from multiple high-quality studies, minimise bias, and increase statistical power. They provide the most reliable summary of available evidence on a question.                                                                                      |
| 13 | Which study design provides the lowest level of evidence?        | <b>Expert opinion / case reports</b>                                                                               | Expert opinion and case reports/series provide the lowest level of evidence because they are based on individual clinical experience or single patient observations, without controlled comparison groups. They are susceptible to bias and cannot establish causality.                                                                                     |
| 14 | What is a pandemic?                                              | <b>A disease outbreak occurring across multiple countries or globally</b>                                          | A pandemic is an epidemic that has spread across multiple countries or continents, affecting a large number of people globally. Examples: COVID-19 (2020), 1918 influenza (Spanish flu), HIV/AIDS. Distinguished from an epidemic (widespread in one region) and endemic (baseline constant presence).                                                      |
| 15 | What is an endemic disease?                                      | <b>A disease consistently present within a particular geographic area or population</b>                            | An endemic disease is one consistently present at a baseline, predictable level within a specific geographic area or population. Examples: malaria in sub-Saharan Africa; dengue in Southeast Asia. Distinguished from epidemic (sudden increase above baseline) and pandemic (global spread).                                                              |
| 16 | What does 'quarantine' mean in public health?                    | <b>Restricting movement of people exposed to a disease to monitor for illness development</b>                      | Quarantine is the restriction of movement of people who have been EXPOSED to a contagious disease but are not yet symptomatic, to monitor whether they develop illness and prevent potential spread. Duration is set to the maximum incubation period of the disease (e.g. 14 days for COVID-19).                                                           |
| 17 | What is the difference between quarantine and isolation?         | <b>Quarantine applies to exposed but asymptomatic individuals; isolation applies to confirmed sick individuals</b> | Quarantine applies to people who are EXPOSED but NOT YET SICK — they are separated to see if they develop disease. Isolation applies to people who are CONFIRMED SICK — they are separated to prevent them from infecting others. Both aim to break the chain of transmission.                                                                              |
| 18 | What is the difference between primary and secondary prevention? | <b>Primary prevention stops disease before onset; secondary detects disease early through screening</b>            | Primary prevention stops disease BEFORE it occurs (e.g. vaccination, health promotion, smoking cessation). Secondary prevention detects disease EARLY to limit harm (e.g. cancer screening, blood pressure checks). Tertiary prevention manages existing disease to reduce complications and disability (e.g. cardiac rehabilitation, diabetes management). |

| #  | Question                                                                    | Answer                                                                                               | Explanation                                                                                                                                                                                                                                                                                                                                           |
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| 19 | What are the social determinants of health?                                 | <b>Non-medical factors such as income, education, housing, and environment that influence health</b> | Social determinants of health (SDOH) are the non-medical conditions in which people are born, grow, live, work, and age. They include: income and wealth, education, employment, housing, access to nutritious food, transport, social support, and the quality of local health services. SDOH are responsible for the majority of health inequities. |
| 20 | How many Sustainable Development Goals (SDGs) are there?                    | <b>17</b>                                                                                            | There are 17 Sustainable Development Goals (SDGs), adopted by all 193 UN member states in September 2015 as part of the 2030 Agenda for Sustainable Development. They replaced the Millennium Development Goals (MDGs) with a broader, universal scope covering 169 targets.                                                                          |
| 21 | What is SDG 1?                                                              | <b>No Poverty</b>                                                                                    | SDG 1: No Poverty - End poverty in all its forms everywhere                                                                                                                                                                                                                                                                                           |
| 22 | What is SDG 2?                                                              | <b>Zero Hunger</b>                                                                                   | SDG 2: Zero Hunger - End hunger, achieve food security and improved nutrition                                                                                                                                                                                                                                                                         |
| 23 | What is SDG 3?                                                              | <b>Good Health and Well-being</b>                                                                    | SDG 3: Good Health and Well-being - Ensure healthy lives and promote well-being for all at all ages                                                                                                                                                                                                                                                   |
| 24 | What is SDG 4?                                                              | <b>Quality Education</b>                                                                             | SDG 4: Quality Education - Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all                                                                                                                                                                                                                       |
| 25 | What is SDG 5?                                                              | <b>Gender Equality</b>                                                                               | SDG 5: Gender Equality - Achieve gender equality and empower all women and girls                                                                                                                                                                                                                                                                      |
| 26 | What is SDG 6?                                                              | <b>Clean Water and Sanitation</b>                                                                    | SDG 6: Clean Water and Sanitation - Ensure availability and sustainable management of water and sanitation for all                                                                                                                                                                                                                                    |
| 27 | What is SDG 13?                                                             | <b>Climate Action</b>                                                                                | SDG 13: Climate Action - Take urgent action to combat climate change and its impacts                                                                                                                                                                                                                                                                  |
| 28 | Which global framework succeeded the Millennium Development Goals (MDGs)?   | <b>The Sustainable Development Goals (SDGs), adopted in 2015</b>                                     | Sustainable Development Goals (SDGs), adopted in 2015                                                                                                                                                                                                                                                                                                 |
| 29 | The Alma-Ata Declaration (1978) is primarily associated with which concept? | <b>Primary Health Care as an approach to achieving Health for All</b>                                | The Alma-Ata Declaration (1978) is the founding document of the Primary Health Care (PHC) movement. It declared that health is a fundamental human right and called for action by all governments to protect and promote health for all people, particularly in low-income countries — leading to the 'Health for All by 2000' campaign.              |
| 30 | What is 'primary health care' as defined at Alma-Ata?                       | <b>Essential health care universally accessible at a cost communities can afford</b>                 | Primary health care (PHC) as defined at Alma-Ata (1978) refers to essential, first-contact health care that is: universally accessible, affordable for communities, delivered as close to home as possible, and covers promotion, prevention, treatment, and rehabilitation. It emphasises community participation and intersectoral collaboration.   |
| 31 | Which disease was declared eradicated by the WHO in 1980?                   | <b>Smallpox — declared eradicated by WHO in 1980</b>                                                 | Smallpox, caused by the Variola virus, was declared eradicated by the WHO in May 1980, following a global vaccination campaign. It is the only human infectious disease to have been completely eradicated. The last naturally occurring case was in 1977 in Somalia.                                                                                 |

| #  | Question                                                                                      | Answer                                                                                      | Explanation                                                                                                                                                                                                                                                                                                                    |
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| 32 | What is the leading cause of death globally?                                                  | <b>Cardiovascular disease (ischaemic heart disease and stroke)</b>                          | Cardiovascular diseases are the leading cause of death globally                                                                                                                                                                                                                                                                |
| 33 | What is the leading cause of death among infectious diseases globally?                        | <b>Tuberculosis (TB)</b>                                                                    | Tuberculosis remains the leading infectious disease killer worldwide                                                                                                                                                                                                                                                           |
| 34 | What is the approximate herd immunity threshold for measles?                                  | <b>95%</b>                                                                                  | ~95%                                                                                                                                                                                                                                                                                                                           |
| 35 | Which country has the highest healthcare expenditure per capita?                              | <b>United States</b>                                                                        | United States has the highest healthcare expenditure per capita                                                                                                                                                                                                                                                                |
| 36 | What is universal health coverage (UHC)?                                                      | <b>All people accessing needed health services without suffering financial hardship</b>     | Universal Health Coverage (UHC) means that all people can access the health services they need — promotive, preventive, curative, rehabilitative, and palliative — without suffering financial hardship. It is a core target of SDG 3. Progress is measured by the UHC Service Coverage Index.                                 |
| 37 | What is the WHO definition of health?                                                         | <b>Complete physical, mental, and social well-being — not merely the absence of disease</b> | The WHO (1948) defines health as: 'a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity.' This holistic definition is important because it acknowledges mental and social dimensions, though it has been criticised for setting an unattainably 'complete' standard. |
| 38 | What is 'health equity'?                                                                      | <b>Absence of unfair and avoidable differences in health among population groups</b>        | Health equity is the absence of unfair, avoidable, or remediable differences in health among population groups defined by social, economic, demographic, or geographic characteristics. It is the goal of health justice — ensuring every person has a fair opportunity to attain their highest level of health.               |
| 39 | Malaria rates rise in a rural district. Which intervention is most likely effective?          | <b>Distribute insecticide-treated bed nets</b>                                              | Insecticide-treated bed nets (ITNs) are one of the most cost-effective, evidence-based interventions for reducing malaria transmission. They provide a physical barrier and kill mosquitoes (Anopheles) that bite at night. Building a hospital addresses treatment, not prevention, which is needed to reduce rising rates.   |
| 40 | A country reports rising child mortality. Which social determinant is most likely implicated? | <b>Lack of clean water and sanitation</b>                                                   | Lack of clean water and sanitation is the most upstream social determinant here. Diarrhoeal diseases from contaminated water are among the leading causes of under-5 mortality globally. Clean water, oral rehydration therapy, and sanitation are more impactful for child mortality than hospital bed counts.                |
| 41 | Why is access to clean water critical for public health?                                      | <b>It prevents waterborne diseases, supports hygiene, and reduces child mortality</b>       | Access to clean water prevents waterborne diseases including cholera, typhoid fever, hepatitis A, and dysentery — which collectively kill millions annually, predominantly children under 5. It also enables hand hygiene (preventing faecal-oral transmission), safe food preparation, and sanitation infrastructure.         |
| 42 | Which combination of interventions best reduces maternal mortality?                           | <b>Skilled birth attendance and emergency obstetric care</b>                                | The most evidence-based combination for reducing maternal mortality is skilled birth attendance (trained midwife/doctor present at delivery) combined with antenatal care (ANC). Emergency obstetric care access, oxytocin for postpartum haemorrhage prevention, and iron/folate supplementation are also critical.           |

| #  | Question                                                                                                                                                                                                                                              | Answer                                                                                                          | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 43 | Which of the following is a key factor contributing to health inequity?                                                                                                                                                                               | <b>Income inequality and limited access to education</b>                                                        | Key contributors to health inequity include: income inequality (poverty limits access to food, housing, healthcare), low educational attainment, gender discrimination, geographic isolation (rural/remote areas), ethnicity and race, inadequate health infrastructure, political instability, and weak governance. These are the structural social determinants of health.                                                                                                                                                                                                                                                                                                                                                 |
| 44 | A study follows 1,000 smokers and 1,000 non-smokers for 10 years. 150 smokers develop lung cancer vs 15 non-smokers. What is the relative risk (RR) of lung cancer in smokers compared to non-smokers?                                                | <b>10</b>                                                                                                       | Relative Risk (RR) = Incidence in exposed group / Incidence in unexposed group. Step 1: Incidence in smokers = $150/1,000 = 0.15$ (15%). Step 2: Incidence in non-smokers = $15/1,000 = 0.015$ (1.5%). Step 3: $RR = 0.15 / 0.015 = 10$ . Interpretation: Smokers are 10 times more likely to develop lung cancer than non-smokers. RR = 1 means no difference; RR > 1 means increased risk in exposed group; RR < 1 means protective effect. RR is used in cohort studies. (Note: Odds Ratio is used in case-control studies.)                                                                                                                                                                                              |
| 45 | A screening test for tuberculosis has 95% sensitivity and 70% specificity. In a population where TB prevalence is 1%, what is the main clinical problem with using this test alone?                                                                   | <b>The low specificity (70%) will generate a large number of false positives in a low-prevalence population</b> | This is the classic low-prevalence screening problem. With 70% specificity, 30% of people WITHOUT TB will test positive (false positives). In a population of 10,000 with 1% TB prevalence: 100 true TB cases, 9,900 without TB. True positives: 95% of 100 = 95. False positives: 30% of 9,900 = 2,970. So for every true positive, there are ~31 false positives. The Positive Predictive Value (PPV) is very low in low-prevalence settings. This is why screening tests need high sensitivity (to catch all cases) AND confirmatory tests with high specificity (to rule out false positives) — the two-step testing strategy.                                                                                           |
| 46 | Country A has an infant mortality rate of 8 per 1,000 live births and spends 11% of GDP on healthcare. Country B has an infant mortality rate of 6 per 1,000 live births but spends only 5% of GDP on healthcare. What does this most likely suggest? | <b>Country B has better social determinants of health</b>                                                       | Country B achieves better health outcomes (lower infant mortality) with less spending — this suggests stronger underlying social determinants of health (SDH): better income equality, higher education levels, stronger sanitation and nutrition, better maternal health, and more equitable access to care. This mirrors real-world comparisons (e.g. Cuba vs USA: Cuba spends far less per capita but has comparable or better infant mortality). It demonstrates that health outcomes are not primarily determined by healthcare spending alone — upstream social, economic, and political factors are often more important. This is the central argument of the WHO Commission on Social Determinants of Health (2008). |
| 47 | A district reports a sudden cluster of 40 cases of acute watery diarrhoea over 3 days, all linked to a single water source. What type of epidemic pattern does this represent?                                                                        | <b>Point source epidemic — all cases linked to a single common exposure at approximately the same time</b>      | A point source epidemic occurs when a group of people are all exposed to the same source at approximately the same time — producing a sharp, rapid rise in cases over a short period (typically within one incubation period). The epidemic curve is characteristically bell-shaped with a steep rise and fall. Classic examples: cholera/typhoid from a contaminated water source, food poisoning from a single meal. This is distinct from a propagated epidemic (person-to-person spread, e.g. measles) which shows a series of peaks over multiple incubation periods. Identifying the epidemic type guides the public health response.                                                                                  |

| #  | Question                                                                                                                                                                                  | Answer                                                                                                            | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 48 | A researcher wants to study the long-term effect of childhood malnutrition on adult cardiovascular disease. Which study design is most appropriate?                                       | <b>Prospective cohort study</b>                                                                                   | A prospective cohort study is the most appropriate design here because: (1) it follows participants FORWARD in time, capturing the temporal relationship between exposure (childhood malnutrition) and outcome (adult CVD); (2) it can measure incidence directly; (3) it allows study of multiple outcomes from one exposure. An RCT would be unethical (cannot randomise children to malnutrition). A case-control study looks backward and cannot establish temporality as clearly. A cross-sectional study measures exposure and outcome at the same time, making it impossible to determine which came first. The Barker hypothesis (developmental origins of health and disease — DOHaD) was established through cohort studies. |
| 49 | Which of the following is the strongest evidence that a risk factor CAUSES a disease (rather than merely being associated with it)?                                                       | <b>Consistent dose-response relationship and biological plausibility replicated across multiple study designs</b> | Causality in epidemiology is assessed using Bradford Hill's Criteria (1965), not any single study type. The strongest evidence for causality requires: (1) Temporal relationship — exposure precedes outcome; (2) Strength of association — large relative risk; (3) Consistency — replicated across studies and populations; (4) Biological plausibility — a credible mechanism exists; (5) Dose-response relationship — more exposure leads to more disease. A single RCT, while gold standard for intervention studies, cannot alone establish causality for a risk factor (often unethical to randomise). Cross-sectional studies cannot establish temporality. Expert opinion is the weakest form of evidence.                    |
| 50 | The maternal mortality ratio (MMR) in Country X is 450 per 100,000 live births. In Country Y it is 12 per 100,000. What is the most likely structural difference between these countries? | <b>Country X has lower access to skilled birth attendance and emergency obstetric care</b>                        | The single strongest predictor of maternal mortality ratio is access to skilled birth attendance (trained midwife or doctor) and emergency obstetric care (EmOC). A MMR of 450/100,000 is characteristic of sub-Saharan African countries with weak health systems; 12/100,000 is typical of high-income countries. The leading causes of maternal death globally — postpartum haemorrhage, eclampsia, sepsis, obstructed labour — are all preventable with skilled attendance and EmOC. Hospital infrastructure spending is secondary; task-shifting to community midwives in low-resource settings has been shown to dramatically reduce MMR even without hospital investment.                                                       |



## Category 4 – Mathematics & Aptitude

| #  | Question                                                                                                         | Answer              | Explanation                                                                                                                                                                                                                                                                                                                                                                                |
|----|------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Find the mean of: 4, 8, 15, 16, 23, 42                                                                           | <b>18</b>           | Mean = sum of all values ÷ number of values. Step 1: Add all values: $4 + 8 + 15 + 16 + 23 + 42 = 108$ . Step 2: Divide by the count (6 numbers): $108 \div 6 = 18$ . Answer: 18. Note: the mean is sensitive to outliers — the 42 pulls it above the middle values.                                                                                                                       |
| 2  | Find the median of: 3, 7, 9, 12, 15, 18, 21                                                                      | <b>12</b>           | Median = the middle value when data is sorted in order. Step 1: Data is already sorted: 3, 7, 9, 12, 15, 18, 21. Step 2: There are 7 values, so the middle is the 4th value. Step 3: Count to the 4th: 3, 7, 9, →12. Answer: 12. The median is unaffected by extreme values.                                                                                                               |
| 3  | Find the median of: 5, 10, 15, 20                                                                                | <b>12.5</b>         | Median with an even number of values = average of the two middle values. Step 1: Data sorted: 5, 10, 15, 20. Step 2: 4 values — middle two are the 2nd and 3rd: 10 and 15. Step 3: Average them: $(10 + 15) \div 2 = 25 \div 2 = 12.5$ . Answer: 12.5.                                                                                                                                     |
| 4  | Find the mode of: 2, 4, 4, 6, 7, 7, 7, 9                                                                         | <b>7</b>            | Mode = the value that appears most frequently. Step 1: List values: 2, 4, 4, 6, 7, 7, 7, 9. Step 2: Count each: 2 appears once, 4 appears twice, 6 once, 7 appears three times, 9 once. Step 3: 7 appears most (3 times). Answer: 7. (There can be no mode, one mode, or multiple modes.)                                                                                                  |
| 5  | Find the range of: 13, 27, 5, 44, 8                                                                              | <b>39</b>           | Range = Maximum value – Minimum value. Step 1: Identify max and min in: 13, 27, 5, 44, 8. Max = 44, Min = 5. Step 2: Range = $44 - 5 = 39$ . Answer: 39. The range is the simplest measure of spread but is highly sensitive to outliers.                                                                                                                                                  |
| 6  | A disease affects 500 people in a population of 10,000. What is the prevalence rate per 1,000?                   | <b>50 per 1,000</b> | Prevalence rate per 1,000 = (number of cases ÷ total population) × 1,000. Step 1: Cases = 500, Population = 10,000. Step 2: $500 \div 10,000 = 0.05$ . Step 3: $0.05 \times 1,000 = 50$ per 1,000. Answer: 50 per 1,000. Common error: dividing 500 by 10 instead of 10,000, giving a false answer of 500 per 1,000. This point prevalence rate helps plan healthcare resource allocation. |
| 7  | A student answers 68 out of 80 questions correctly. What is their percentage score?                              | <b>85%</b>          | Percentage score = (correct answers ÷ total questions) × 100. Step 1: $68 \div 80 = 0.85$ . Step 2: $0.85 \times 100 = 85\%$ . Answer: 85%.                                                                                                                                                                                                                                                |
| 8  | The mean of five numbers is 14. Four of them are 10, 12, 16, 18. What is the fifth number?                       | <b>14</b>           | Step 1: Find the total sum needed. Mean = 14, number of values = 5. Total = $14 \times 5 = 70$ . Step 2: Sum the four known values: $10 + 12 + 16 + 18 = 56$ . Step 3: Fifth number = Total – Known sum = $70 - 56 = 14$ . Answer: 14.                                                                                                                                                     |
| 9  | What is the probability of getting heads AND rolling a 6 on a 6-sided die simultaneously?                        | <b>1/12</b>         | For independent events (one does not affect the other), multiply their probabilities. $P(\text{heads}) = 1/2$ . $P(\text{rolling a 6}) = 1/6$ . $P(\text{both}) = 1/2 \times 1/6 = 1/12 \approx 0.083$ (or ~8.3%). Answer: 1/12.                                                                                                                                                           |
| 10 | What is the probability of a specific outcome when tossing a 24-sided die, 6-sided die, and coin simultaneously? | <b>1/288</b>        | For three independent events, multiply all three probabilities. $P(\text{specific face on 24-sided die}) = 1/24$ . $P(\text{specific face on 6-sided die}) = 1/6$ . $P(\text{specific side of coin}) = 1/2$ . $P(\text{all three specific outcomes}) = 1/24 \times 1/6 \times 1/2 = 1/288 \approx 0.0035$ . Answer: 1/288. This question appeared in the actual DIT exam.                  |

| #  | Question                                                                                         | Answer              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 11 | A bag has 3 red, 5 blue, and 2 green balls. What is the probability of picking a blue ball?      | <b>1/2</b>          | $P(\text{blue}) = \text{number of blue balls} \div \text{total balls}$ . Blue = 5, Total = $3 + 5 + 2 = 10$ . $P(\text{blue}) = 5 \div 10 = 1/2 = 0.5$ . Answer: 1/2 (50%). Each ball is assumed equally likely to be picked.                                                                                                                                                                                                        |
| 12 | What is the probability of NOT rolling a 1 on a 6-sided die?                                     | <b>5/6</b>          | Use the complement rule: $P(\text{event does NOT happen}) = 1 - P(\text{event happens})$ . $P(\text{rolling a 1}) = 1/6$ . $P(\text{NOT rolling a 1}) = 1 - 1/6 = 5/6 \approx 0.833$ . Answer: 5/6. The complement rule is faster than counting all other outcomes (2, 3, 4, 5, 6 = 5 faces out of 6). Note: 6/7 is not a valid probability from a 6-sided die — always check your denominator matches the total number of outcomes. |
| 13 | Solve: $5x + 3 = 3x + 11$                                                                        | <b>4</b>            | $5x + 3 = 3x + 11$ . Step 1: Move x terms to one side: $5x - 3x = 11 - 3$ . Step 2: $2x = 8$ . Step 3: $x = 8 \div 2 = 4$ . Answer: $x = 4$ . Check: $5(4) + 3 = 23$ ; $3(4) + 11 = 23$ ✓                                                                                                                                                                                                                                            |
| 14 | Solve: $4x - 2 = 2x + 8$                                                                         | <b>5</b>            | $4x - 2 = 2x + 8$ . Step 1: Move x terms to one side: $4x - 2x = 8 + 2$ . Step 2: $2x = 10$ . Step 3: $x = 10 \div 2 = 5$ . Answer: $x = 5$ . Check: $4(5) - 2 = 18$ ; $2(5) + 8 = 18$ ✓                                                                                                                                                                                                                                             |
| 15 | Solve: $3(2x - 1) = 21$                                                                          | <b>4</b>            | $3(2x - 1) = 21$ . Step 1: Expand: $6x - 3 = 21$ . Step 2: Add 3 to both sides: $6x = 24$ . Step 3: Divide by 6: $x = 4$ . Answer: $x = 4$ . Check: $3(2 \times 4 - 1) = 3(7) = 21$ ✓                                                                                                                                                                                                                                                |
| 16 | Solve: $(x + 2)/3 = 5$                                                                           | <b>13</b>           | $(x + 2)/3 = 5$ . Step 1: Multiply both sides by 3: $x + 2 = 15$ . Step 2: Subtract 2 from both sides: $x = 13$ . Answer: $x = 13$ . Check: $(13 + 2)/3 = 15/3 = 5$ ✓                                                                                                                                                                                                                                                                |
| 17 | Solve: $7x - 4 = 3x + 16$                                                                        | <b>5</b>            | $7x - 4 = 3x + 16$ . Step 1: Move x terms to one side: $7x - 3x = 16 + 4$ . Step 2: $4x = 20$ . Step 3: $x = 20 \div 4 = 5$ . Answer: $x = 5$ . Check: $7(5) - 4 = 31$ ; $3(5) + 16 = 31$ ✓                                                                                                                                                                                                                                          |
| 18 | A hospital reduces waiting time by 15%. Original wait was 40 minutes. What is the new wait time? | <b>34 min</b>       | New wait time = Original $\times (1 - \text{reduction}\%)$ . Step 1: Reduction = $15\% = 0.15$ . Step 2: New time = $40 \times (1 - 0.15) = 40 \times 0.85 = 34$ minutes. Answer: 34 minutes. Alternatively: 15% of 40 = 6 minutes saved; $40 - 6 = 34$ .                                                                                                                                                                            |
| 19 | A vaccine is 90% effective. In 2,000 vaccinated people, how many may still get the disease?      | <b>200</b>          | If a vaccine is 90% effective, then 10% of vaccinated people may still contract the disease. Step 1: 10% of 2,000 = $2,000 \times 0.10 = 200$ . Answer: 200 people. This illustrates that no vaccine is 100% effective — this is why herd immunity also matters.                                                                                                                                                                     |
| 20 | A country spends 6% of its \$500 billion GDP on healthcare. How much is that?                    | <b>\$30 billion</b> | Healthcare spending = GDP $\times$ percentage. Step 1: 6% of \$500 billion = $500 \times 0.06 = \$30$ billion. Answer: \$30 billion. Context: WHO recommends low-income countries spend at least 5% of GDP on health to achieve universal health coverage.                                                                                                                                                                           |
| 21 | Population grows from 8 million to 9.2 million. What is the percentage increase?                 | <b>15%</b>          | Percentage increase = $(\text{increase} \div \text{original}) \times 100$ . Step 1: Increase = $9.2 - 8.0 = 1.2$ million. Step 2: Percentage increase = $(1.2 \div 8.0) \times 100 = 0.15 \times 100 = 15\%$ . Answer: 15%.                                                                                                                                                                                                          |
| 22 | Drug A costs \$250. Drug B costs 40% more. What does Drug B cost?                                | <b>\$350</b>        | Drug B = Drug A cost $\times (1 + \text{increase}\%)$ . Step 1: 40% more than \$250 = $250 \times 1.40 = \$350$ . Answer: \$350. Alternatively: 40% of 250 = 100; $250 + 100 = 350$ .                                                                                                                                                                                                                                                |

| #  | Question                                                                                                                                                                                                | Answer                                                                              | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 23 | 3 out of every 1,000 people develop a condition annually. What is the incidence rate per 100,000?                                                                                                       | <b>300 per 100,000</b>                                                              | Incidence rate per 100,000 = (cases ÷ population) × 100,000. Step 1: $3 \div 1,000 = 0.003$ . Step 2: $0.003 \times 100,000 = 300$ per 100,000 per year. Answer: 300 per 100,000. Expressing rates per 100,000 is the standard in epidemiology for comparability across populations.                                                                                                                                                                                                                                                                                                                |
| 24 | All doctors are graduates. Some graduates are researchers. What can we conclude?                                                                                                                        | <b>Some doctors may be researchers, but this cannot be concluded with certainty</b> | From the premises: ALL doctors are graduates, and SOME graduates are researchers. We cannot conclude that any doctors are researchers — the two groups (doctors and researchers) may not overlap at all. The answer is that some doctors MAY be researchers, but this CANNOT be concluded with certainty. This is a fallacy of the undistributed middle.                                                                                                                                                                                                                                            |
| 25 | If cats can swim and birds can fly, which statement MUST be true?                                                                                                                                       | <b>Cats can swim</b>                                                                | From the premises: cats CAN swim (stated), birds CAN fly (stated). Option A (cats can fly) — NOT stated. Option B (birds can swim) — NOT stated. Option C (cats can swim) — directly stated. Option D (birds cannot swim) — NOT stated. Answer: C. Only what is directly given in the premises can be concluded.                                                                                                                                                                                                                                                                                    |
| 26 | A is taller than B. C is shorter than A but taller than B. Who is the shortest?                                                                                                                         | <b>B</b>                                                                            | Order the heights using the clues: $A > C$ (C is shorter than A) and $C > B$ (C is taller than B). Therefore the order is: A (tallest) > C (middle) > B (shortest). Answer: B is the shortest.                                                                                                                                                                                                                                                                                                                                                                                                      |
| 27 | A patient receives a drug every 8 hours. First dose at 6:00 AM. When is the third dose?                                                                                                                 | <b>10:00 PM</b>                                                                     | Count forward in 8-hour intervals from the first dose: 1st dose: 6:00 AM. 2nd dose: 6:00 AM + 8 hours = 2:00 PM. 3rd dose: 2:00 PM + 8 hours = 10:00 PM. Answer: 10:00 PM. This type of medication timing question tests basic schedule arithmetic.                                                                                                                                                                                                                                                                                                                                                 |
| 28 | Some nurses are managers. All managers attend meetings. What can we conclude?                                                                                                                           | <b>Some nurses attend meetings — specifically those who are managers</b>            | From the premises: SOME nurses are managers, and ALL managers attend meetings. The nurses who are managers attend meetings (because all managers do). But we cannot say ALL nurses attend meetings — only those who are managers do. Answer: Some nurses attend meetings — True; but we CANNOT conclude all nurses do.                                                                                                                                                                                                                                                                              |
| 29 | A dataset has values: 5, 7, 7, 8, 9, 10, 12, 15, 15, 52. Which measure of central tendency best represents this dataset and why?                                                                        | <b>Median = 9.5; it is not affected by the extreme outlier of 52</b>                | Step 1 — Calculate mean: $\text{Sum} = 5+7+7+8+9+10+12+15+15+52 = 140$ . $\text{Mean} = 140/10 = 14$ . But 52 is a clear outlier — it pulls the mean above 9 of the 10 data points. Step 2 — Calculate median: Sort (already sorted). 10 values, so median = average of 5th and 6th values = $(9+10)/2 = 9.5$ . Step 3 — The median (9.5) better represents the 'typical' value since 9 out of 10 data points are at or below 15. The mean (14) is distorted by the outlier 52. Rule: Use median when data is skewed or contains outliers. Use mean for symmetrical distributions without outliers. |
| 30 | A clinic's costs are modelled as $C = 500 + 12n$ , where C is total cost in dollars and n is the number of patients. If the budget is \$2,900, what is the maximum number of patients that can be seen? | <b>200 patients</b>                                                                 | Solve for n when $C = 2,900$ . Step 1: $2,900 = 500 + 12n$ . Step 2: $2,900 - 500 = 12n \rightarrow 2,400 = 12n$ . Step 3: $n = 2,400 / 12 = 200$ . Answer: 200 patients. This type of linear cost model is used in health economics and clinic planning. The '500' represents fixed costs (overheads, staff salaries) and '12n' represents variable costs (consumables, time per patient). Understanding the relationship between fixed costs, variable costs, and patient volume is fundamental to healthcare budgeting.                                                                          |

| #  | Question                                                                                                                                                                                              | Answer                                            | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| 31 | A vaccination programme costs \$2.4 million and prevents 800 DALYs. A competing programme costs \$1.8 million and prevents 450 DALYs. Which programme is more cost-effective?                         | <b>Programme 1 — \$3,000 per DALY averted</b>     | Cost-effectiveness = Cost / DALYs averted (cost per DALY averted). Programme 1: \$2,400,000 / 800 DALYs = \$3,000 per DALY averted. Programme 2: \$1,800,000 / 450 DALYs = \$4,000 per DALY averted. Programme 1 is more cost-effective because it costs LESS per DALY averted (\$3,000 vs \$4,000). Important: total cost and total benefit are not the criteria — the RATIO matters. The WHO threshold for cost-effectiveness is approximately 1-3 times the GDP per capita per DALY averted. Cost per DALY is a standard metric in global health resource allocation (e.g. used by the Bill & Melinda Gates Foundation and GAVI). |
| 32 | A country's population is 25 million. The crude birth rate is 32 per 1,000 and the crude death rate is 8 per 1,000. What is the natural population growth rate per year, and how many people is that? | <b>2.4% growth — 600,000 people per year</b>      | Natural growth rate = Crude Birth Rate - Crude Death Rate. Step 1: 32 - 8 = 24 per 1,000 per year = 2.4% per year. Step 2: Absolute growth = 2.4% × 25,000,000 = 0.024 × 25,000,000 = 600,000 people per year. Answer: 2.4% growth rate — 600,000 additional people per year. This natural increase does not account for migration. At 2.4% annual growth, the population doubles in approximately 29 years (Rule of 70: doubling time = 70 / growth rate%). Understanding population growth rates is fundamental to public health planning — it determines future demand for healthcare, education, water, and food.                |
| 33 | In a town, 70% of people drink coffee and 40% drink tea. 20% drink both. What percentage drink neither coffee nor tea?                                                                                | <b>10%</b>                                        | Use the inclusion-exclusion principle: P(coffee OR tea) = P(coffee) + P(tea) - P(both). = 70% + 40% - 20% = 90%. Therefore P(neither) = 100% - 90% = 10%. Answer: 10% drink neither. The -20% prevents double-counting people who drink both. This principle applies in public health when calculating the proportion of a population affected by at least one of multiple risk factors or conditions (e.g. what % have hypertension OR diabetes OR both).                                                                                                                                                                           |
| 34 | A health worker can vaccinate 18 children per hour. A clinic session runs from 8:00 AM to 1:00 PM with a 30-minute break. How many children can be vaccinated in one session?                         | <b>81</b>                                         | Step 1: Calculate working time. Session: 8:00 AM to 1:00 PM = 5 hours total. Less 30-minute break = 4.5 hours of working time. Step 2: Children vaccinated = 18 per hour × 4.5 hours = 81 children. Answer: 81 children. This type of workforce productivity calculation is used in immunisation programme planning, outreach campaign scheduling, and health worker capacity assessment. Knowing session capacity helps determine how many sessions are needed to achieve coverage targets in a given population.                                                                                                                   |
| 35 | A dataset of patient waiting times (minutes) is: 12, 15, 15, 18, 20, 22, 22, 22, 30, 44. What is the median waiting time?                                                                             | <b>21 minutes (average of 5th and 6th values)</b> | Median with an even number of values = average of the two middle values. Step 1: Data is already sorted: 12, 15, 15, 18, 20, 22, 22, 22, 30, 44. Step 2: 10 values — middle two are the 5th and 6th values: 20 and 22. Step 3: Median = (20 + 22) / 2 = 42 / 2 = 21 minutes. Answer: 21. Note: the mean would be (12+15+15+18+20+22+22+22+30+44)/10 = 220/10 = 22, pulled slightly higher by the outlier 44. The median (21) is a better representation of the typical wait.                                                                                                                                                         |
| 36 | A public health team records the number of malaria cases per week over 6 weeks: 14, 18, 22, 16, 20, 18. What is the mean number of weekly cases, and what is the range?                               | <b>Mean = 18, Range = 8</b>                       | Step 1 — Mean: Add all values: 14 + 18 + 22 + 16 + 20 + 18 = 108. Divide by 6: 108 / 6 = 18. Step 2 — Range: Maximum = 22, Minimum = 14. Range = 22 - 14 = 8. Answer: Mean = 18, Range = 8. In disease surveillance, mean weekly cases indicates the average burden, while range shows how much transmission fluctuates week to week — useful for spotting outbreak peaks.                                                                                                                                                                                                                                                           |

| #  | Question                                                                                                                                                                                              | Answer                                           | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 37 | A nutrition survey of 5 children records weights (kg): 12, 14, 16, 14, 19. What are the mean, median, and mode?                                                                                       | <b>Mean = 15, Median = 14, Mode = 14</b>         | Step 1 — Mean: Sum = 12+14+16+14+19 = 75. Mean = 75/5 = 15 kg. Step 2 — Median: Sort the values: 12, 14, 14, 16, 19. Middle value (3rd of 5) = 14 kg. Step 3 — Mode: 14 appears twice (most frequent). Mode = 14 kg. Answer: Mean = 15, Median = 14, Mode = 14. Note that the mean (15) is pulled up by the highest value (19), while the median and mode both reflect the lower cluster — showing why multiple measures of central tendency are useful in health data.                                                                                                                                                                                                                                                       |
| 38 | A clinic tests patients for two independent conditions. The probability of testing positive for condition A is 0.3, and for condition B is 0.4. What is the probability of testing positive for BOTH? | <b>0.12 (multiply the probabilities)</b>         | For two independent events, $P(A \text{ AND } B) = P(A) \times P(B)$ . $P(A) = 0.3$ , $P(B) = 0.4$ . $P(\text{both}) = 0.3 \times 0.4 = 0.12$ . Answer: 0.12 (12%). This is the multiplication rule for independent events — the outcome of one test does not affect the other. Common error: adding the probabilities ( $0.3 + 0.4 = 0.7$ ) gives $P(A \text{ OR } B)$ , not $P(A \text{ AND } B)$ . The union formula for OR is: $P(A \text{ OR } B) = P(A) + P(B) - P(A \text{ AND } B) = 0.3 + 0.4 - 0.12 = 0.58$ .                                                                                                                                                                                                       |
| 39 | A bag contains 4 red, 6 blue, and 10 green balls. What is the probability of NOT picking a green ball?                                                                                                | <b>0.50 — P(not green) = 1 – P(green) = 0.50</b> | Step 1: Total balls = 4 + 6 + 10 = 20. Step 2: $P(\text{green}) = 10/20 = 0.5$ . Step 3: Use the complement rule: $P(\text{NOT green}) = 1 - P(\text{green}) = 1 - 0.5 = 0.5$ . Answer: 0.5. Non-green balls = 4 red + 6 blue = 10 out of 20, confirming $P(\text{not green}) = 10/20 = 0.5$ . The complement rule ( $P(\text{not } A) = 1 - P(A)$ ) is often the fastest approach when there are many outcomes to count.                                                                                                                                                                                                                                                                                                     |
| 40 | A health centre's monthly budget is \$B. After spending \$1,200 on salaries and a third of the remainder on supplies, \$800 is left. What was the original budget?                                    | <b>\$2,400</b>                                   | Work backwards from what's left. After salaries: remaining = $B - 1,200$ . One third of remainder spent on supplies, so two thirds remain: $(2/3)(B - 1,200) = 800$ . Multiply both sides by 3/2: $B - 1,200 = 800 \times 3/2 = 1,200$ . Therefore $B = 1,200 + 1,200 = \$3,000$ . Answer: \$3,000. Check: Start with \$3,000. After salaries: \$1,800 left. One third on supplies = \$600. Remaining = \$1,800 - \$600 = \$1,200... Let us re-verify: $(2/3)(3000-1200) = (2/3)(1800) = 1200 \neq 800$ . Correcting: $(2/3)(B-1200)=800 \rightarrow B-1200=1200 \rightarrow B=2400$ . Check: $(2/3)(2400-1200)=(2/3)(1200)=800 \checkmark$ . Correct answer is \$2,400 — option A. Note: always verify by substituting back. |
| 41 | A vaccination team needs to cover 3,600 households. Team A can visit 45 households per day and Team B can visit 30 households per day. Working together, how many days will it take?                  | <b>48 days</b>                                   | Combined rate = 45 + 30 = 75 households per day. Days needed = $3,600 / 75 = 48$ days. Answer: 48 days. This is a combined work rate problem. The key step is adding the rates (not the times). Check: Team A in 48 days: $45 \times 48 = 2,160$ . Team B in 48 days: $30 \times 48 = 1,440$ . Total: $2,160 + 1,440 = 3,600 \checkmark$ . This type of calculation is used in planning immunisation campaigns, outreach schedules, and field survey timelines.                                                                                                                                                                                                                                                               |
| 42 | A country has 50,000 births per year. The under-5 mortality rate is 60 per 1,000 live births. How many children under 5 die each year?                                                                | <b>3,000 children per year</b>                   | Deaths = (rate per 1,000 / 1,000) $\times$ number of live births. Step 1: Rate = 60 per 1,000 = 0.06 (6%). Step 2: Deaths = $0.06 \times 50,000 = 3,000$ children per year. Answer: 3,000. Alternatively: $60/1,000 \times 50,000 = 60 \times 50 = 3,000$ . Under-5 mortality rate (U5MR) is one of the most important indicators of child health and overall development — a U5MR of 60/1,000 is typical of a lower-middle-income country. The global average is approximately 37/1,000 (2023).                                                                                                                                                                                                                              |

| #  | Question                                                                                                                                                    | Answer                                                                                                     | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| 43 | In a health survey, the mean systolic blood pressure of a group is 125 mmHg with a standard deviation of 10 mmHg. What does the standard deviation tell us? | <b>Most readings fall between 115 and 135 mmHg — showing how spread out the values are around the mean</b> | Standard deviation (SD) measures the spread of data around the mean. An SD of 10 mmHg means most values (approximately 68% in a normal distribution) fall within one SD of the mean — i.e. between $125 - 10 = 115$ mmHg and $125 + 10 = 135$ mmHg. A small SD means values are clustered tightly around the mean; a large SD means they are widely spread. SD does NOT mean the average is 10 or that the max is 135 — it describes variability, not a boundary.                                                                            |
| 44 | Two clinics measure patient waiting times. Clinic A: mean = 20 min, SD = 2 min. Clinic B: mean = 20 min, SD = 15 min. What does this tell us?               | <b>Clinic A has more consistent waiting times; Clinic B is far more variable</b>                           | Both clinics have the same mean waiting time (20 minutes), but very different standard deviations. Clinic A (SD = 2) has very consistent times — most patients wait close to 20 minutes. Clinic B (SD = 15) is highly variable — some patients may wait only 5 minutes while others wait 35 minutes or more. The mean alone does not tell the full story; SD reveals the reliability and consistency of a service. In public health, high variability in care delivery often indicates inequity or system inefficiency.                      |
| 45 | A dataset of children's heights (cm) has a mean of 110 cm and SD of 5 cm. Approximately what percentage of children fall between 100 cm and 120 cm?         | <b>95%</b>                                                                                                 | Using the 68-95-99.7 rule (Empirical Rule) for normal distributions: 68% of values fall within 1 SD of the mean (105–115 cm here). 95% fall within 2 SDs of the mean. 99.7% fall within 3 SDs. Here: $100 \text{ cm} = 110 - 2 \times 5 = \text{mean} - 2 \text{ SD}$ . $120 \text{ cm} = 110 + 2 \times 5 = \text{mean} + 2 \text{ SD}$ . So the range 100–120 cm represents $\text{mean} \pm 2 \text{ SD} \rightarrow$ approximately 95% of children. This rule is essential for interpreting population health data and reference ranges. |
| 46 | Which of the following statements about standard deviation is TRUE?                                                                                         | <b>SD measures how spread out values are around the mean</b>                                               | Standard deviation (SD) measures the dispersion or spread of data values around the mean. Option A is FALSE — a higher SD means more variability, not more accuracy; in fact high variability can indicate inconsistency. Option B is FALSE — SD = 0 when all values are identical (e.g. everyone scored exactly 50). Option D is FALSE — mean and SD are independent measures; they can be any values. SD is one of the most important descriptive statistics in public health and clinical research.                                       |